

Isolation of normal canine mast cell progenitors, melanocytes and keratinocytes as within-patient comparators for tumour transcriptomics

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Abstract

Precision oncology asks which genes are altered in a particular patient's tumour, and answering that by transcriptome sequencing requires a matched population of the normal cell the tumour came from. For three common canine tumours (melanoma, mast cell tumour and squamous cell carcinoma), that normal cell is rare in its home tissue: melanocytes are 3-5% of epidermal cells, cutaneous mast cells roughly 0.1%, and circulating mast cell progenitors 0.1-0.5% of blood leucocytes. Four isolation routes were built and tested. Enzymatic digestion of canine skin yielded 2.2×10^6 cells/mL but no cell with mast cell morphology, so mast cells were instead pursued as blood-borne progenitors. Magnetic depletion of CD90⁺ cells followed by magnetic CD117 selection raised the CD117⁺ fraction from 0.23% to 1.92%, but left 73% of the sample double-negative and no histamine-granulated cells on cytology. Magnetic CD90 depletion followed by fluorescence-activated sorting instead returned a discrete CD117⁺CD90⁻ population of 8.3×10^3 cells, 0.188% of input, matching the reported progenitor frequency and sufficient for single-cell library preparation. A dispase-based epidermal peel gave abundant pigmented melanocytes from oral mucosa but only ~1 melanocyte per field from haired flank skin, where hair follicles anchored the epidermis; the same protocol with keratinocyte medium gave 1.5×10^6 keratinocytes with minor melanocyte carry-over. Anagen hair follicle explants produced no outgrowth. No route gave a pure culture, but the keratinocyte, oral-mucosal melanocyte and blood progenitor preparations are each compatible with single-cell sequencing, where purity is replaced by marker-based selection after capture.

Keywords: mast cell progenitor · melanocyte · keratinocyte · CD117/c-Kit · flow cytometry · canine melanoma · mast cell tumour · single-cell RNA sequencing

1. Introduction

Two patients can present with histologically identical tumours, at the same stage, receive the same therapy and do quite differently. Conventional cytotoxic chemotherapy and surgery address this heterogeneity by not addressing it: treatment is either broad and toxic or invasive, and local recurrence or metastasis is common. Precision oncology proposes instead to read each tumour's transcriptome and select targets from it. That comparison is only interpretable against a baseline: the transcriptome of the normal cell type that gave rise to the tumour, ideally from the same patient.

The dog is an informative model here because several of its spontaneous tumours are both common and mechanistically close to the human disease. Canine melanoma runs to upwards of 100,000 cases per year and is predominantly oral or mucosal rather than sun-driven, so unlike human cutaneous melanoma it rarely carries *BRAF* or *NRAS* hot-spot mutations while still showing PI3K pathway activation [1,2]. Cutaneous mast cell tumours make up 7-25% of canine skin tumours, can kill through degranulation-driven anaphylaxis or gastroduodenal perforation, and are frequently driven by activating mutations in CD117 (c-Kit) [3]. Squamous cell and basal cell carcinoma arise from keratinocytes in both species.

The obstacle is upstream of sequencing: obtaining the normal comparator at all. Each target cell is a small minority of its tissue, each sits in a matrix of more abundant and faster-growing neighbours, and each must be captured with minimal time in culture, because extended culture selects subclones and shifts expression away from the *in vivo* state the comparison depends on. This work therefore developed and evaluated isolation protocols for canine mast cells (or their progenitors), melanocytes and keratinocytes, judged not by culture convenience but by whether the output is usable as a normal comparator in transcriptome sequencing.

2. Materials and methods

2.1 Tissue and blood sources

Skin biopsies were taken from necropsy dogs dead no longer than 24 h and held at 4 °C. Hair was removed with unguarded clippers followed by a disposable razor, and 14 mm punch biopsies were taken from the flank; oral mucosa was cut as 5 × 5 mm squares. Samples travelled on ice in Ca²⁺/Mg²⁺-free Hank's balanced salt solution with penicillin (300 U/mL), streptomycin (300 µg/mL) and amphotericin B (0.75 µg/mL) for no longer than 1 h. Blood (40 mL) was drawn from the jugular vein into EDTA under an approved Auburn University College of Veterinary Medicine IACUC protocol.

2.2 Skin mast cell digestion

After defatting, 1.06 g of finely chopped skin was digested for 180 min at 37 °C with shaking in 15 mL DMEM containing collagenase I (34.5 mg), hyaluronidase type I-S (18 mg), protease type XIV (12 mg) and bovine albumin fraction V (0.45 g), the enzyme mixture of Kawarai *et al.* [4]. The digest was filtered at 100 µm, centrifuged at 400 × g for 10 min, resuspended in DMEM + 5% FBS with glutamine and antibiotics, and cultured at 37 °C, 5% CO₂ for no more than 5 days.

2.3 PBMC isolation, antibody panel

Blood was diluted 1:1 in PBS + 2% FBS, layered over 15 mL Histopaque 1077 in SepMate tubes and centrifuged at 1200 × g for 10 min; the mononuclear fraction was washed twice and residual erythrocytes lysed in two rounds of ACK buffer. PE anti-CD117 (clone 2B8) and APC anti-dog CD90 (clone YKIX337.217) were titrated by flow cytometry across 0.025-1.25 µg on lines of known phenotype: the canine mast cell tumour line MPT-1, normal canine fibroblasts (NCF), fetal dog kidney (FDK) cells and PBMC.

2.4 Magnetic separation and cell sorting

PBMC were blocked in 10% normal dog serum with anti-dog Fc receptor for 1 h, stained with 0.125 µg of each antibody for 1 h and washed. CD90⁺ cells were depleted on an APC positive-selection kit with magnetic nanoparticles. Two downstream routes were then compared. **Route A** continued with a releasable PE positive-selection kit to enrich CD117⁺ cells magnetically, with every wash fraction analysed by flow cytometry to quantify target-cell loss. **Route B** instead sorted the CD90-depleted fraction on a MoFlo XDP high-speed sorter, gating sequentially on live cells, singlets, CD117⁺ and CD90⁻; sorted cells and the aborted fraction were both re-analysed. Analytical cytometry used a CytoFLEX LX. Route A product was also cytocentrifuged at 800 rpm for 3 min and stained with modified Wright's Giemsa for histamine granules, against MPT-1 as a positive control.

2.5 Melanocyte and keratinocyte isolation

Biopsies were washed in 70% ethanol, defatted, cut into small sections and incubated overnight at 4 °C in a Petri dish in CnT-09 keratinocyte isolation medium containing 10 mg/mL dispase grade II. Samples were warmed to 37 °C, the epidermis peeled from the dermis with forceps, sliced, and dispersed in 1× TryPLE for 15 min with vortexing. The suspension was filtered at 100 µm, washed, and plated either in Melanocyte Growth Medium M2 (melanocytes) or CnT-09 (keratinocytes, a protocol adapted from Sandey *et al.* [10]) and followed for two to three weeks. Growth medium omitted both the phorbol ester TPA and cholera toxin: Hsu *et al.* [5] omit TPA but retain cholera toxin, and this protocol departs from theirs in dropping it too. In an amended version the dispersed epidermal suspension was stained with near-UV viability dye and anti-CD117 and analysed directly by cytometry instead of cultured.

2.6 Hair follicle explants

About 100 dark hairs were plucked from the craniodorsal region; 30-50 were confirmed anagen at 10× magnification by bulb size and integrity, trimmed to ~2 cm, washed 20 min in antibiotic-supplemented M2 medium, and explanted 12 per well onto 0.4 µm microporous membranes, following Dieckmann *et al.* [6]. Cultures were followed for three to six weeks.

3. Results

3.1 Antibody panel behaved as required

CD90 saturated at the recommended 0.125 µg, labelled 60-80% of PBMC and was positive on NCF but negative on MPT-1 and FDK; CD117 also saturated at 0.125 µg, positive on MPT-1 and negative on NCF. The panel therefore separates a mast-cell-lineage marker from a marker of fibroblasts and thymocyte-lineage cells. Both results were confirmed in one replicate.

3.2 Mast cells could not be recovered from skin

Digestion of 1.06 g of flank skin gave 2.2×10^6 cells/mL. Cells adhered within 24 h and by day 3 the culture was dominated by spindle-shaped, fibroblast-like cells. No cell with mast cell phenotype (highly granular, round to ovoid, lymphocyte-sized, with a well-formed round nucleus) was seen in any field examined.

3.3 Two routes to a blood CD117⁺ population

One PBMC preparation yielded 6.355×10^7 cells at 91% viability. Magnetic CD90 depletion reduced this to 4.4×10^6 cells and cut CD90 positivity by 45.12%. Magnetic CD117 selection (Route A) returned 4.635×10^4 cells and raised the CD117⁺ fraction from

0.23% in the stained, unseparated control to 1.92% (Fig. 1), an eight-fold enrichment but with 73.06% of events still negative for both markers. Wash-fraction analysis localised the losses: the first CD90 depletion wash was almost purely CD90⁺, as intended, but the second discarded 0.3% PE⁺ and 44% double-negative cells, and the CD117 selection washes discarded PE⁺ cells at 1.41% and 1.52%. Cytospins of the Route A product were heterogeneous, with some granulated cells but none matching the MPT-1 control morphology.

Route B performed better. Of all logged events, 3.92% were CD90⁺ and 0.16% fell in the CD117⁺CD90⁻ gate. Sorting recovered 8.3×10^3 cells, 0.188% of the 4.4×10^6 input, and post-sort re-analysis confirmed a discrete CD117⁺CD90⁻ population. That frequency sits at the reported 0.1% blood abundance of mast cell progenitors, and the yield exceeds the input requirement for single-cell library preparation at 2000 cells/ μ L.

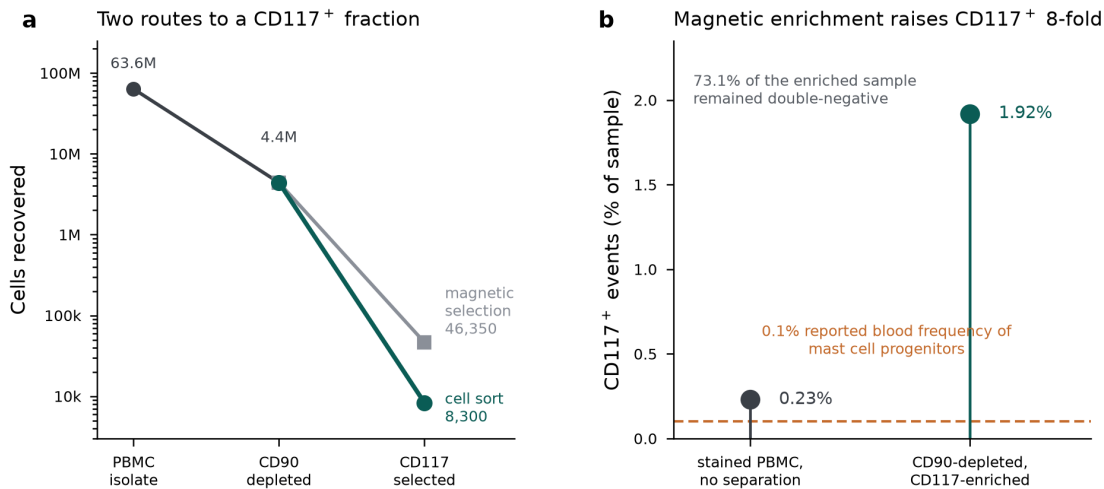


Figure 1. Recovery and enrichment along the two blood separation routes. (a) Cells recovered at each step. Both routes share a common magnetic CD90 depletion (dark grey) before diverging into magnetic CD117 selection (light grey) or fluorescence-activated sorting (teal). (b) CD117⁺ events as a percentage of the sample, measured by flow cytometry on the stained but unseparated control and on the Route A product; the dashed line marks the published blood frequency of mast cell progenitors. Higher is better in (b). One donor preparation ($n = 1$); percentages are of total analysed events.

3.4 Source tissue decided the epidermal outcome

Oral mucosa dissociated readily and yielded many large, multipolar, heavily pigmented melanocytes, intermingled with transparent ovoid keratinocytes that persisted and multiplied despite melanocyte-selective medium. Haired flank skin behaved differently: only 1.53×10^6 cells were recovered, the flask held mostly debris, and melanocytes appeared at roughly one per field of view. Staining dispersed epidermis for CD117 directly, to bypass culture altogether, was inconclusive: autofluorescence from cornified stratum corneum debris obscured any CD117⁺ population. The same peel run into keratinocyte medium gave 1.5×10^6 cells with many keratinocytes per field and a minority of melanocytes visibly less pigmented than in M2 medium. Anagen follicle explants produced no cellular outgrowth over six weeks.

Protocol	Source	Yield	Phenotype obtained	Fit for comparator use
Enzymatic skin digest	Flank skin, 1.06 g	2.2×10^6 cells/mL	Spindle, fibroblast-like; no histamine-granulated cells	No: target absent
Route A: CD90 depletion + magnetic CD117	40 mL blood	4.635×10^4 cells; 1.92% CD117 ⁺	Heterogeneous, 73% double-negative; no histamine granules	Marginal: purity too low
Route B: CD90 depletion + cell sort	40 mL blood	8.3×10^3 cells; 0.188% of input	Discrete CD117 ⁺ CD90 ⁻ population	Yes: in canine patient use
Dispase peel, melanocyte medium	Oral mucosa	Abundant pigmented cells	Multipolar melanocytes + keratinocytes	Yes, for single-cell work
Dispase peel, melanocyte medium	Flank skin	1.53×10^6 cells; ~1 melanocyte/field	Debris-dominated; few melanocytes	No: insufficient yield
Dispase peel, keratinocyte medium	Flank skin	1.5×10^6 cells	Many keratinocytes; few pale melanocytes	Yes: in canine patient use
Anagen follicle explant	Craniodorsal hair	None	No outgrowth in 6 weeks	No: protocol failed

Table 1. Outcome of each isolation route. Fitness is judged against the intended use (a normal comparator population for single-cell transcriptome sequencing), not against culture viability. Yields are from single preparations.

4. Discussion

Skin is the wrong compartment for mast cells. Kawarai *et al.* [4] obtained mast cells from canine skin, but only after five weeks of culture. Using their enzyme mixture directly, no mast-cell-phenotype cell appeared here. Beyond the failure to replicate, the five-week requirement is itself disqualifying: a population that needs a month of culture to emerge has had ample opportunity to drift from the expression state it is meant to represent. Moving to blood was therefore not only a workaround but an improvement in principle, since circulating progenitors [9] can be captured the same day.

Sorting beats magnetic selection when the target is very rare. Route A's eight-fold enrichment sounds useful until the composition is examined: at 1.92% CD117⁺ with 73% double-negative cells, most sequencing reads would be spent on bystanders. The recovered cell number ($3\text{--}4.6 \times 10^4$) also exceeds what a 0.1%-abundant population should give from this input, which is itself evidence of carry-over rather than selection. Route B cost more instrument time but returned a population whose frequency, 0.188%, matches the expected progenitor abundance. The wash audit explains part of the difference (magnetic selection discarded PE⁺ cells at every wash) and argues that for a target at this frequency the binary capture of a sorter is worth its overhead.

The population is enriched, not identified. CD117 is rare enough in blood that mast cell progenitors should make up a substantial share of a CD117⁺CD90⁻ gate, but nothing here demonstrates that they do. Confirmation requires transcript-level evidence: *KIT*, histidine decarboxylase (*Hdc*), the FcεRI β chain (*Ms4a2*), α- and β-tryptase (*Tpsab1*, *Tpsb2*) and carboxypeptidase A3 (*Cpa3*). Adding FcεRI to the sort panel would sharpen the gate: in human blood, Lin⁻CD117⁺FcεRI⁺ cells gave rise to mature mast cells at 75.3% efficiency versus 4.4% for their FcεRI⁻ counterparts, and already express mast cell genes at the progenitor stage [7]. A progenitor lacking histamine granules is an imperfect stand-in for a mature tissue mast cell, but mast cell tumour cells are themselves dedifferentiated, so the progenitor may be the closer developmental match. Which comparator is appropriate is an empirical question the sequencing can answer.

Hair follicles are a mechanical problem, not an enzymatic one. Flank epidermis tore away in chunks rather than sheets because follicles anchored it to the dermis; dispase concentration and digestion time were not limiting. Razor shaving and moving the overnight digestion from a conical tube to a Petri dish (improving enzyme access across the flat specimen) both helped, and hairless oral mucosa avoided the problem entirely. Bao *et al.* [8] ran a comparable protocol in arctic fox without reporting how they handled follicular anchoring. Melanocyte yield from haired canine skin is thus limited by how much epidermis is lost with the follicles, which is why oral mucosa gave the better culture.

Medium choice trades purity against normality. Melanocyte protocols conventionally use TPA with cholera toxin to promote melanocyte growth while suppressing keratinocytes, and even the phorbol-free protocol of Hsu *et al.* [5] retains cholera toxin. Both are deliberate perturbations (TPA is a tumour promoter acting through protein kinase C, cholera toxin an intracellular cAMP enhancer), and both alter transcription in exactly the cells whose untouched transcriptome is the object of the experiment. Omitting them preserved normality at the cost of purity: keratinocytes persisted and proliferated in M2 medium, and melanocytes persisted in keratinocyte medium, confirming that neither commercial medium is genuinely selective.

Single-cell sequencing changes what purity is for. None of these preparations is pure, which would have been fatal for bulk RNA sequencing, where two cell types' transcriptomes blend irreversibly. In single-cell workflows, purification only has to concentrate the target enough to be found: melanocytes can be picked out of a mixed epidermal culture by tyrosinase expression and contaminating profiles dropped from the analysis. On that criterion the keratinocyte, oral-mucosal melanocyte and Route B progenitor preparations all qualify; flank-skin melanocytes and the follicle explants do not.

Limitations. The quantitative flow and yield data come from single preparations ($n = 1$ donor for the blood work; titrations confirmed in one replicate), so no variance is estimated and the enrichment figures are demonstrations rather than performance characteristics. Identity of the sorted CD117⁺CD90⁻ population as mast cell progenitors is inferred from surface phenotype alone; the confirmatory single-cell sequencing was not performed. Tissue came from dogs dead up to 24 h, so some cells had probably entered senescence or apoptosis before processing. The number of cultures and fields examined for the skin mast cell digest was not recorded in the source document, so that negative result is qualitative.

5. Conclusions

Of four isolation routes, two are in use with canine patients: magnetic CD90 depletion followed by fluorescence-activated sorting for blood mast cell progenitors, and the dispase epidermal peel for keratinocytes. Oral mucosa is a productive melanocyte source; haired flank skin is not, and plucked anagen follicles yielded nothing. No route produced a pure population, but purity is the wrong specification for single-cell sequencing, which selects target cells computationally after capture. Two problems remain open. The

sorted progenitor population needs transcript-level confirmation of mast cell identity, best combined with FcεRI in the sort panel. And melanocyte comparators for oral melanoma need a surgically acceptable source: surgeons resecting an oral tumour are reluctant to create a second mucosal wound, so sampling the tumour margin would solve the access problem and raise a new one (establishing that margin melanocytes are actually normal). That question, too, is answerable by single-cell sequencing.

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